KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY

ASSIGNMENT – 1

U21CS501 – WEB TECHNOLOGIES

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III CSE – D

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1. Scenario: A newly established educational institution plans to launch its official website. Before

developing the website, the management wants to understand the fundamental concepts of the

World Wide Web.

Task: Create a simple HTML webpage explaining the following concepts with suitable illustrations

or diagrams:

- **Internet Overview**
- **Fundamental Computer Network Concepts**
- **URL and its Structure**
- **Domain Name**
- **Web Browsers and Web Servers**
- **Working Principle of a Website**
- **Difference between Client-side and Server-side Scripting**

<u>Program:</u>

<u>index.html:</u>

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>World Wide Web</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <!-- Header section -->
    <header>
        <h1>World Wide Web</h1>
        <p>Understanding Web Technology</p>
    </header>
    <!-- Navigation menu -->
```

```html
    <nav>
        <a href="#internet">Internet</a>
        <a href="#url">URL</a>
        <a href="#browser">Browser</a>
    </nav>
    <!-- Main content -->
    <main>
        <section id="internet">
            <h2>Internet</h2>
            <p>The Internet connects computers and devices worldwide.</p>
        </section>
        <section id="url">
            <h2>URL</h2>
            <p>A URL identifies the location of a web resource.</p>
        </section>
        <section id="browser">
            <h2>Web Browser</h2>
            <p>A browser is used to access and display webpages.</p>
        </section>
    </main>
    <!-- Footer -->
    <footer>
        <p>Web Technology Assignment</p>
    </footer>
</body>
</html>
```

Style.css:

```css
* {
    box-sizing: border-box;
```

```
    margin: 0;
    padding: 0;
}
body {
    font-family: Arial, sans-serif;
    background: #f4f7fb;
    color: #222;
    line-height: 1.6;
}
header {
    background: #2563eb;
    color: white;
    text-align: center;
    padding: 30px;
}
nav {
    background: #111827;
    text-align: center;
    padding: 12px;
}
nav a {
    color: white;
    text-decoration: none;
    margin: 15px;
}
nav a:hover {
    color: #60a5fa;
}
main {
    width: 90%;
    margin: 25px auto;
```

```
}

section {
    background: white;
    padding: 25px;
    margin-bottom: 20px;
    border-radius: 10px;
}
h2 {
    color: #2563eb;
    margin-bottom: 10px;
}
footer {
    background: #111827;
    color: white;
    text-align: center;
    padding: 20px;
}
```

Output:

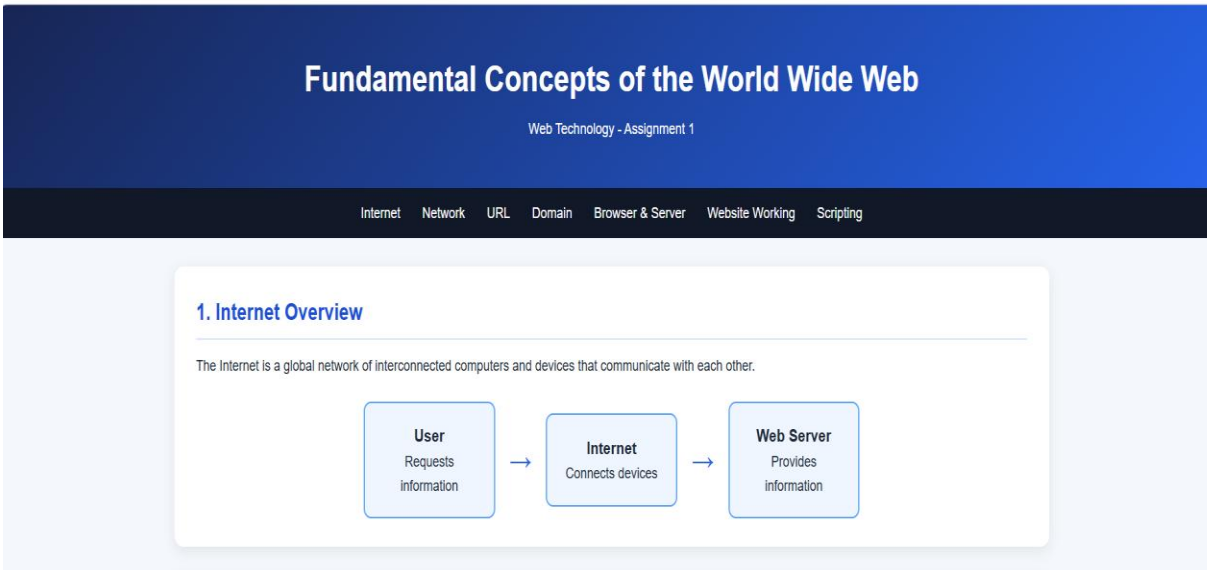


Figure no : 1 – Internet Overview

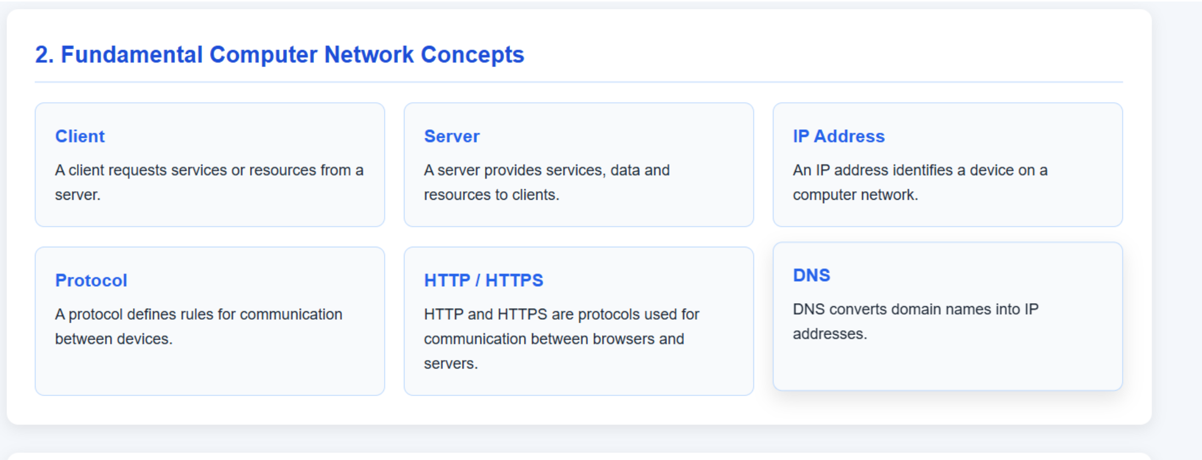


Figure no: 2 – Fundamental Computer Network Concepts

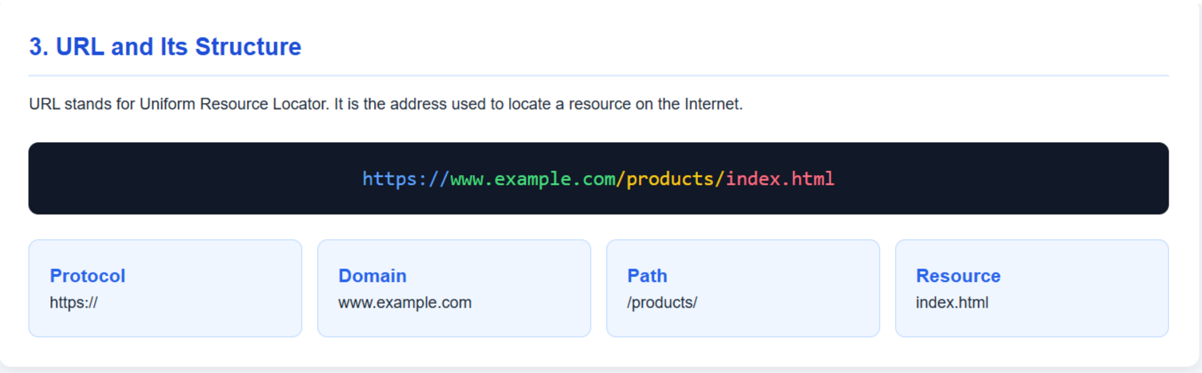


Figure no : 3 – URL and its Structure

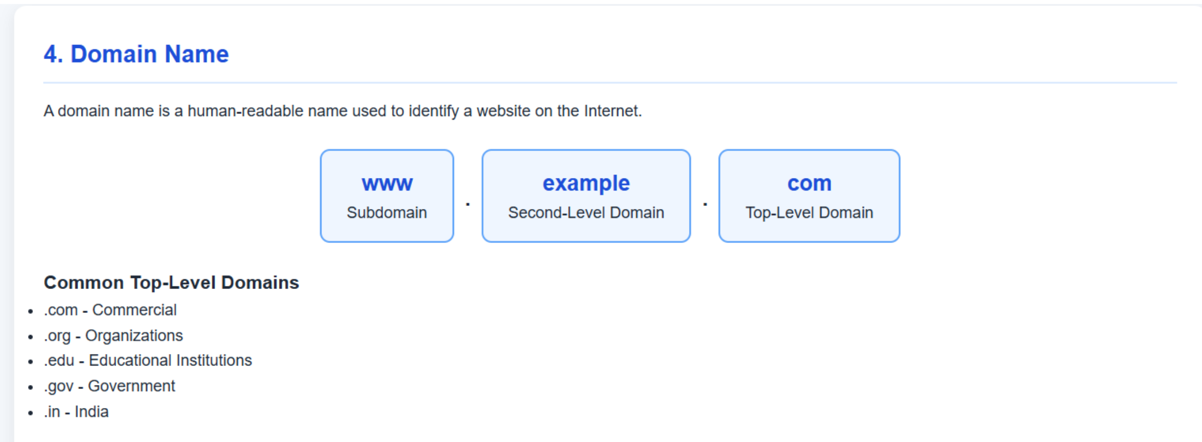


Figure no : 4 – Domain Name

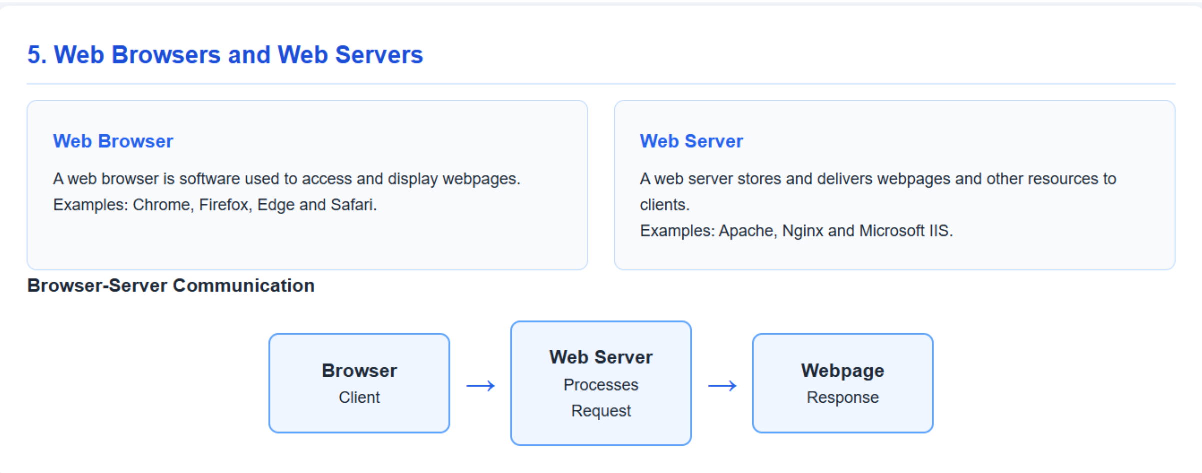


Figure no : 5 – Web Browsers and Web Servers

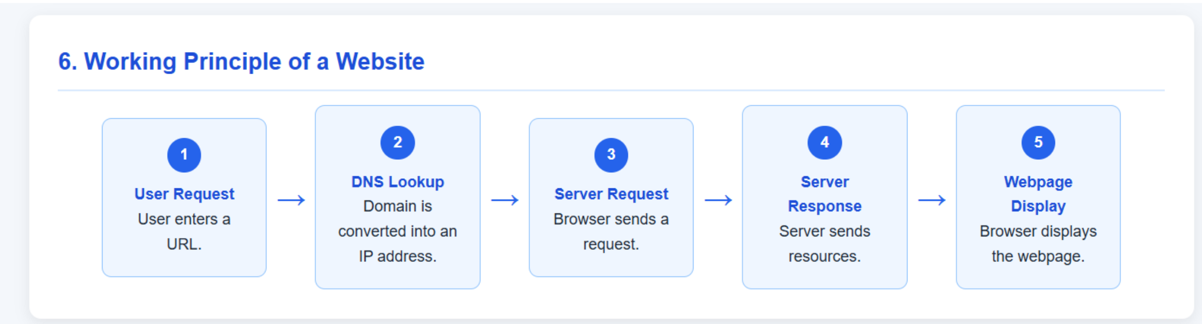
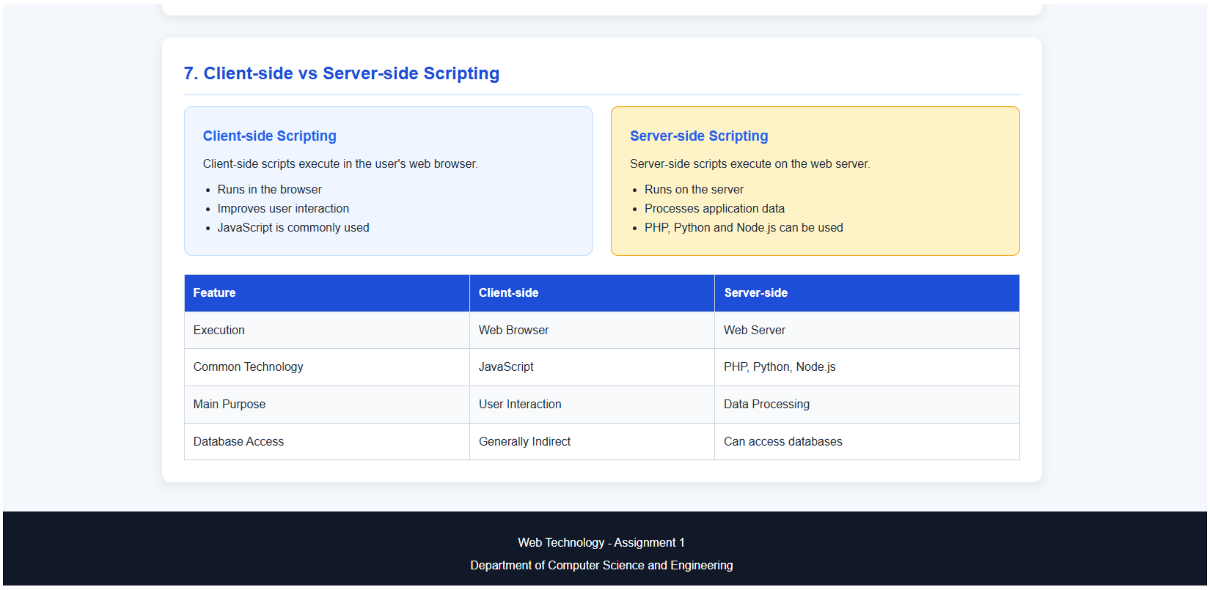


Figure no : 6 – Working Principle of a website



Feature	Client-side	Server-side
Execution	Web Browser	Web Server
Common Technology	JavaScript	PHP, Python, Node.js
Main Purpose	User Interaction	Data Processing
Database Access	Generally Indirect	Can access databases

Figure no : 7 – Clident – side vs Server-side Scripting

Explanation of Code:

- The <nav> element creates navigation links to different sections of the webpage.
- The <main> element contains the main content of the webpage.
- The <section> elements organize the content into different topics such as Internet, URL, and Web Browser.
- The <h1> and <h2> elements are used to display the main heading and section headings.
- The <p> element is used to display descriptive information about each topic.
- The <a> elements create clickable navigation links between different sections.
- The <link> tag connects the HTML page with the external CSS file.
- The <footer> section displays the assignment and webpage information at the bottom of the page.
- CSS is used to style the webpage by adding colors, spacing, backgrounds, borders, and layouts.
- The nav CSS rules are used to design the navigation menu and its links.
- The section CSS rules are used to create organized content areas with backgrounds, padding, margins, and rounded corners.
- The :hover property is used to create an interactive effect when the user moves the mouse over navigation links.
- Media queries are used in CSS to make the webpage responsive on different screen sizes.
- Comments such as <!-- Header section --> and <!-- Navigation menu --> are included in the HTML code to clearly identify different parts of the program.

Conclusion:

The webpage was successfully designed using **HTML5 and CSS3** to present the fundamental concepts of the World Wide Web. It includes navigation links, organized sections, suitable layouts, and basic responsive design. The webpage provides a clear understanding of Internet, URL, browser, server, and other web technology concepts.

2. Scenario: A college wants to digitize its student admission process.

Task: Design a responsive Student Admission Form using HTML5 and CSS3. Your webpage should
demonstrate:

- **HTML Form Elements**
- **Different Input Types**
- **Media Elements (Image/Video/Audio)**
- **CSS3 Selectors**
- **Box Model**
- **Responsive User Interface**

Program:

index.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Student Admission Form</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>
   <h1>Student Admission Form</h1>
   <img src="college.jpg" alt="College Campus">
   <form>
      <label>Name:</label>
      <input type="text" required>
      <label>Email:</label>
      <input type="email" required>
      <label>Gender:</label>
      <input type="radio" name="g"> Male
      <input type="radio" name="g"> Female
      <label>Course:</label>
      <select><option>CSE</option><option>AIML</option></select>
      <label>Address:</label>
```

```
        <textarea></textarea>
        <input type="file">
        <button type="submit">Submit</button>
    </form>
    <h2>College Media</h2>
    <audio controls></audio>
    <video controls><source src="college-video.mp4"></video>
</body>
</html>
```

Style.css

```
* { box-sizing: border-box; }

body {
    font-family: Arial;
    background: #eef4f8;
    margin: 20px;
}
h1, h2 {
    color: #1769aa;
    text-align: center;
}
form {
    max-width: 600px;
    margin: auto;
    padding: 20px;
    background: white;
}
img, video, audio {
    width: 100%;
    margin: 10px 0;
}
```

```css
input, select, textarea {
    width: 100%;
    padding: 8px;
    margin: 8px 0;
}
button {
    padding: 10px 20px;
    background: #1769aa;
    color: white;
    border: none;
}
button:hover {
    background: #0d4d7d;
}
@media (max-width: 600px) {
    body { margin: 10px; }
}
```

Output:

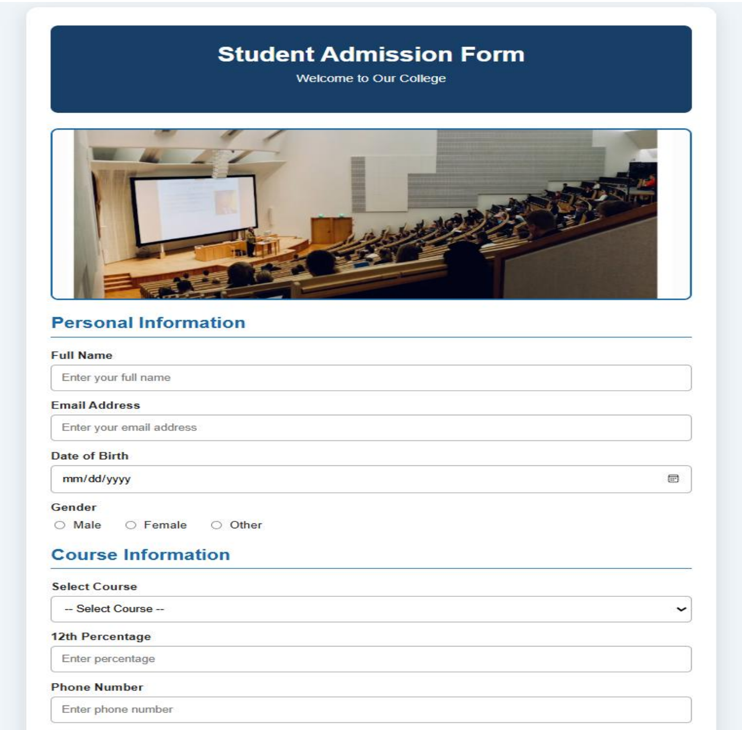


Figure no : 8 – Student Admission Form

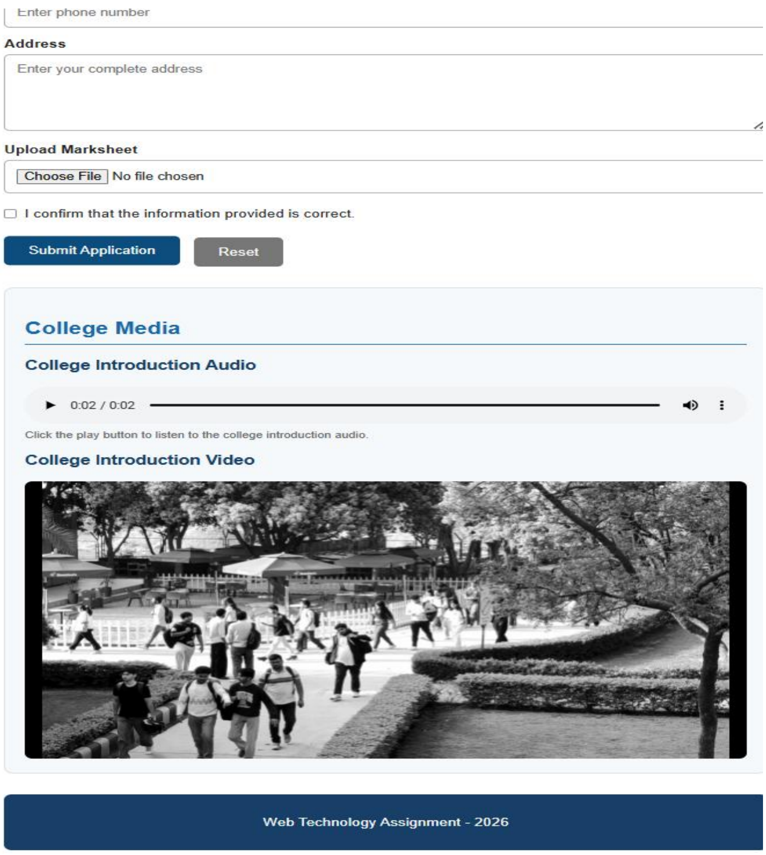


Figure no : 9 – College Media

Explanation of the Code:

- The <form> element creates the student admission form.
- Text, email, date, number, radio, checkbox, file, select and textarea controls collect different types of information.
- The <img> tag displays the college campus image.
- The <audio> and <video> elements provide multimedia content.
- The external style.css file is connected using the <link> tag.
- CSS controls the layout, colours, spacing, borders and fonts.
- :focus and :hover are CSS selectors used for interactive effects.
- margin, padding, border and width demonstrate the CSS box model.
- The @media rule makes the webpage responsive on smaller screens.

Conclusion:

The Student Admission Form successfully demonstrates the use of **HTML form elements, multimedia elements and CSS3 styling**. Different input controls make the form interactive, while CSS improves its appearance and layout. The responsive design allows the webpage to adapt to different screen sizes. Thus, the webpage provides a simple and user-friendly college admission interface.

3. Scenario: An online fashion store wants an attractive home page to improve customer engagement.

Task: Design the homepage using HTML5 and CSS3 by incorporating:

- **Backgrounds and Borders**
- **Text Effects**
- **CSS Animations**
- **Multiple Column Layout**
- **User Interface Design Principles**

Program

Index.html:

```
<!DOCTYPE html>
<html>
<head>
  <!-- PAGE TITLE -->
  <title>StyleAura Fashion Store</title>
  <!-- CONNECT CSS FILE -->
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <!-- HEADER SECTION -->
  <header>
    <h1>STYLEAURA</h1>
    <p>Fashion That Defines You</p>
  </header>
```

```html
    <!-- NAVIGATION SECTION -->
    <nav>
        <a href="#home">Home</a>
        <a href="#collections">Collections</a>
        <a href="#offers">Offers</a>
    </nav>
    <!-- HERO SECTION -->
    <section id="home">
        <h2>Find Your Style</h2>
        <p>Discover the latest fashion trends.</p>
        <button>Shop Now</button>
    </section>
    <!-- COLLECTION SECTION -->
    <section id="collections">
        <h2>Our Collections</h2>
        <div class="cards">
            <div>Men's Fashion</div>
            <div>Women's Fashion</div>
            <div>Footwear</div>
            <div>Accessories</div>
        </div>
    </section>
    <!-- OFFER SECTION -->
    <section id="offers">
        <h2>Up to 50% OFF</h2>
    </section>
    <!-- FOOTER SECTION -->
    <footer>
        <p>© 2026 StyleAura</p>
    </footer>
</body>
</html>
```

Style.css:

```
/* GENERAL STYLING */
body {
   margin: 0;
   font-family: Arial;
   background: #f8f6f4;
}
/* HEADER SECTION */
header {
   text-align: center;
   padding: 30px;
   background: #171717;
   color: white;
}
/* NAVIGATION SECTION */
nav {
   text-align: center;
   padding: 15px;
   background: white;
}
nav a {
   margin: 15px;
   text-decoration: none;
}
/* HERO SECTION */
#home {
   text-align: center;
   padding: 80px;
   background: #c89b6d;
   color: white;
}
/* MULTIPLE-COLUMN LAYOUT */
.cards {
```

```
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 15px;
    padding: 30px;
}
/* HOVER EFFECT */
.cards div:hover {
    transform: translateY(-5px);
}
/* FOOTER SECTION */
footer {
    text-align: center;
    padding: 25px;
    background: #171717;
    color: white;
}
```

Output:

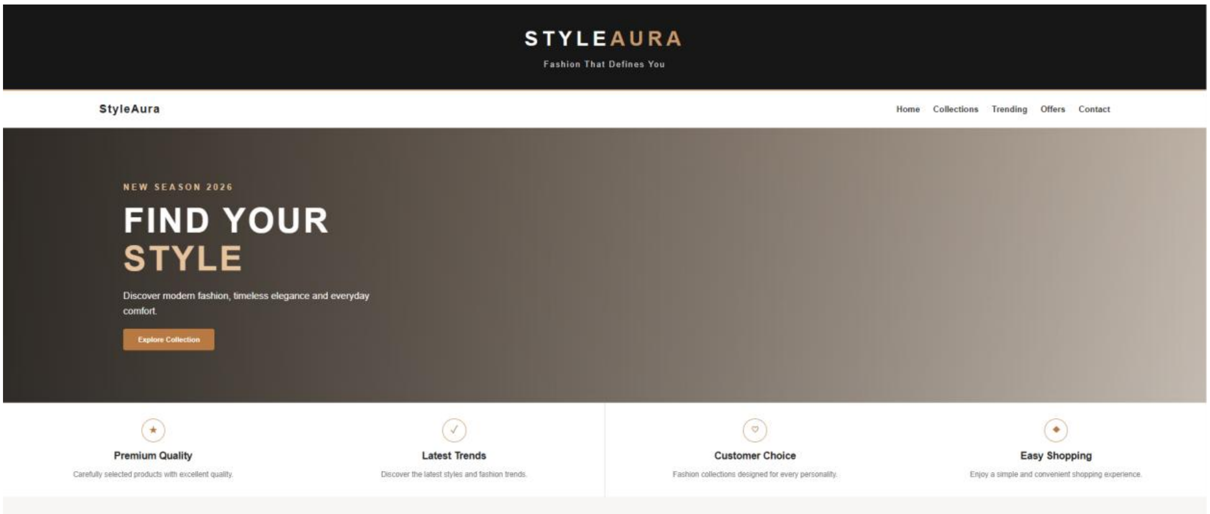


Figure no : 10 – Home page

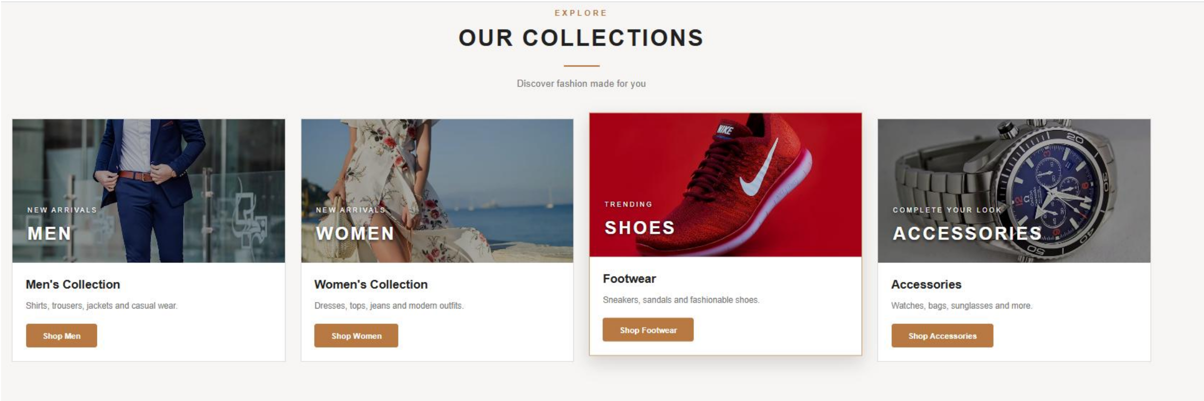


Figure no : 11 – Collection Page

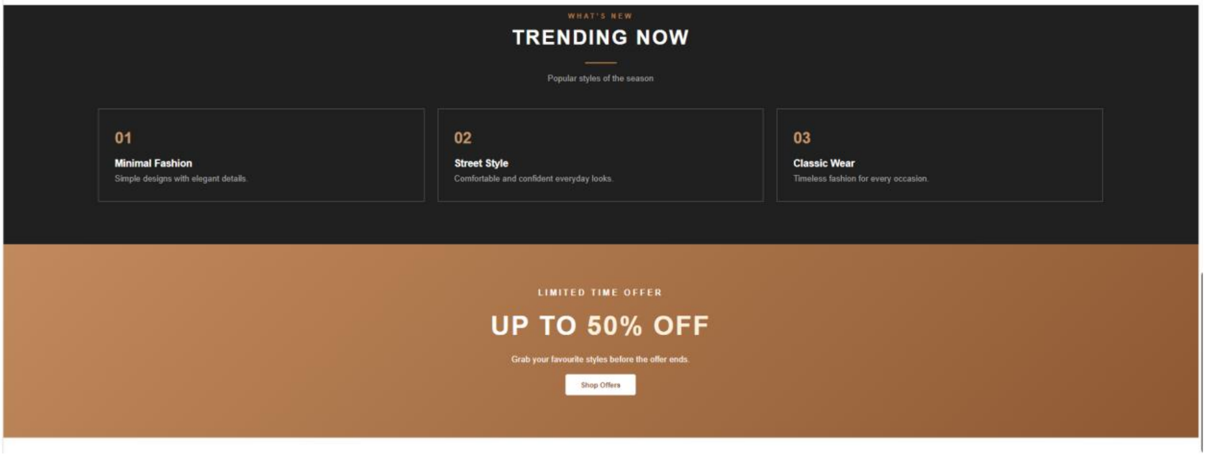


Figure no : 12 – Trendings and offer Page

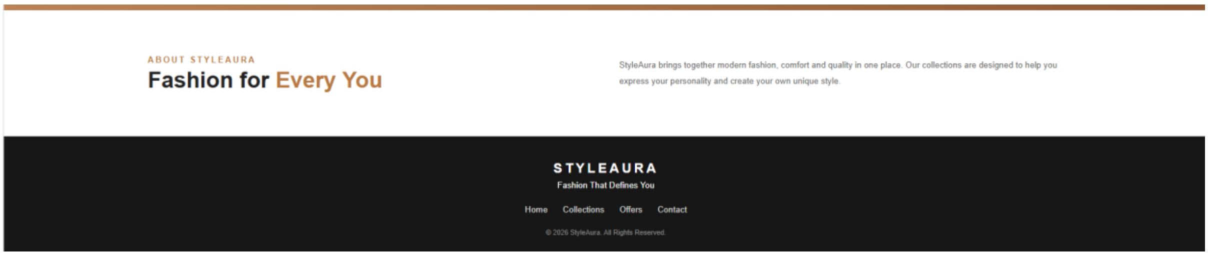


Figure no : 13 – About and contact Page

Explanation of Code:

- The <header> section displays the **StyleAura fashion store name and tagline**.
- The <nav> element creates navigation links such as **Home, Collections, Offers and Contact**.
- The <section> elements divide the webpage into different content areas.
- The **hero section** displays the main fashion message and a **Shop Now** button.
- The **collections section** displays categories such as **Men, Women, Footwear and Accessories**.
- The <img> tag is used to display fashion images for the different collections.
- CSS **Grid** is used to arrange the collection cards in multiple columns.
- CSS backgrounds, borders and box shadows are used to improve the appearance of the webpage.
- **Hover effects and CSS animations** provide interactive visual effects.
- The @media rule makes the webpage **responsive on different screen sizes**.
- The <footer> section displays the copyright and website information.

Conclusion

The StyleAura fashion store webpage demonstrates the use of **HTML5 and CSS3** to create an attractive and responsive website. It includes navigation, images, multiple-column layout, backgrounds, borders, hover effects and CSS animations.

4.Scenario: A school wants to automate the student grade calculation process.

Task: Develop a JavaScript program that accepts marks of five subjects and displays the Total, Average, Grade, and Pass/Fail status. The solution should demonstrate:

• Variables and Data Types

• Literals

• Operators

• Statements (Selection and Iteration)

• User-defined Functions

Program:

Index.html:

```html
<!DOCTYPE html>
<html>
<head>
   <!-- PAGE TITLE -->
   <title>Student Grade Calculator</title>
   <!-- CONNECT CSS FILE -->
   <link rel="stylesheet" href="style.css">
</head>
<body>
   <!-- MAIN CONTAINER -->
   <div class="container">
      <!-- PAGE HEADING -->
      <h1>Student Grade Calculator</h1>
      <!-- STUDENT NAME -->
      <input type="text" id="studentName"
            placeholder="Enter Student Name">
      <!-- SUBJECT MARKS -->
      <input type="number" id="m1" placeholder="Subject 1">
      <input type="number" id="m2" placeholder="Subject 2">
      <input type="number" id="m3" placeholder="Subject 3">
      <input type="number" id="m4" placeholder="Subject 4">
      <input type="number" id="m5" placeholder="Subject 5">
      <!-- ACTION BUTTONS -->
      <button onclick="calculate()">Calculate</button>
      <button onclick="resetForm()">Reset</button>
      <!-- RESULT DISPLAY -->
      <div id="result">
         Result will appear here.
      </div>
```

```
    </div>
    <!-- CONNECT JAVASCRIPT FILE -->
    <script src="script.js"></script>
</body>
</html>
```

Style.css:

```
/* ================= GENERAL STYLING ================= */
body {
    font-family: Arial, sans-serif;
    background: #eef4f8;
    text-align: center;
    padding: 40px;
}
/* ================= MAIN CONTAINER ================= */
.container {
    width: 400px;
    margin: auto;
    padding: 30px;
    background: white;
    border-radius: 12px;
    box-shadow: 0 5px 15px #aaa;
}
/* ================= HEADING ================= */
h1 {
    color: #1769aa;
}
/* ================= INPUT FIELDS ================= */
input {
    width: 90%;
    padding: 10px;
    margin: 7px;
    border: 1px solid #aaa;
```

```
    border-radius: 5px;
}
/* ================== BUTTONS ================== */
button {
    padding: 10px 20px;
    margin: 10px 5px;
    border: none;
    border-radius: 5px;
    background: #1769aa;
    color: white;
    cursor: pointer;
}
/* ================== HOVER EFFECT ================== */
button:hover {
    background: #0d4f82;
}
/* ================== RESULT ================== */
#result {
    margin-top: 20px;
    padding: 15px;
    background: #f2f8fc;
    line-height: 1.8;
}
```

Script.js:

```
// ================== CALCULATE FUNCTION ==================
function calculate() {
    // GET STUDENT NAME
    let name = document.getElementById("studentName").value;
    // GET SUBJECT MARKS
    let a = Number(document.getElementById("m1").value);
    let b = Number(document.getElementById("m2").value);
    let c = Number(document.getElementById("m3").value);
```

```
    let d = Number(document.getElementById("m4").value);
    let e = Number(document.getElementById("m5").value);
    // CALCULATE TOTAL
    let total = a + b + c + d + e;
    // CALCULATE AVERAGE
    let average = total / 5;
    // DECLARE GRADE
    let grade;
    // DETERMINE GRADE
    if (average >= 90)
        grade = "A+";
    else if (average >= 80)
        grade = "A";
    else if (average >= 70)
        grade = "B";
    else if (average >= 60)
        grade = "C";
    else
        grade = "F";
    // DETERMINE PASS OR FAIL
    let status;
    if (a >= 40 && b >= 40 && c >= 40 &&
        d >= 40 && e >= 40)
        status = "PASS";
    else
        status = "FAIL";
    // DISPLAY RESULT
    document.getElementById("result").innerHTML =
        "Name: " + name + "<br>" +
        "Total: " + total + "<br>" +
        "Average: " + average.toFixed(2) + "<br>" +
        "Grade: " + grade + "<br>" +
        "Status: " + status;
}
```

```
// ================== RESET FUNCTION ==================
function resetForm() {
    // CLEAR ALL INPUTS
    document.getElementById("studentName").value = "";
    document.getElementById("m1").value = "";
    document.getElementById("m2").value = "";
    document.getElementById("m3").value = "";
    document.getElementById("m4").value = "";
    document.getElementById("m5").value = "";
    // CLEAR RESULT
    document.getElementById("result").innerHTML =
        "Result will appear here.";
}
```

Output:

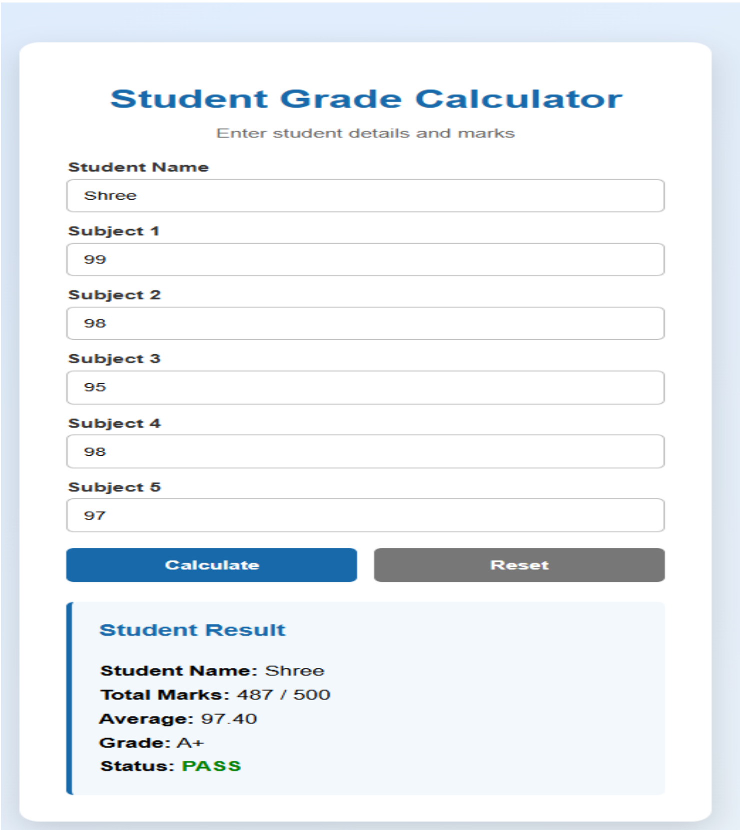


Figure no : 14 – Student Grade Calculator

Explanation of Code:

- The <input> elements accept the **student name and marks of five subjects**.
- CSS is used to style the calculator using **background, borders, spacing and hover effects**.
- JavaScript let variables store the student details and marks.
- Number() converts the entered marks into numeric values.
- Arithmetic operators calculate the **total and average**.
- if...else if...else statements determine the **grade**.
- The && operator checks whether the student has scored at least **40 marks in every subject**.
- The user-defined calculate() function performs the calculation and displays the result.
- The resetForm() function clears the entered values and result.

Conclusion:

The Student Grade Calculator demonstrates the use of **HTML, CSS and JavaScript** to create an interactive webpage. It accepts student details and subject marks and calculates the **Total, Average, Grade and Pass/Fail status**. The program also demonstrates variables, data types, literals, operators, selection statements and user-defined functions as required by the question.

5.Scenario: A supermarket wants to generate customer bills automatically.
Task: Develop a JavaScript program to calculate the bill amount for three purchased products. The
program should:
• Accept quantity and unit price for each product
• Calculate the total bill
• Apply a 10% discount if the total amount exceeds ₹2000
• Display the final payable amount
The solution should demonstrate:
• Variables and Data Types
• Literals
• Operators
• Statements
• User-defined Functions

Program:

Index.html:

```html
<!DOCTYPE html>
<html>
<head>
   <!-- PAGE TITLE -->
   <title>Supermarket Billing System</title>
   <!-- CONNECT CSS FILE -->
   <link rel="stylesheet" href="style.css">
</head>
<body>
   <!-- MAIN CONTAINER -->
   <div class="container">
      <!-- PAGE HEADING -->
      <h1>Supermarket Billing System</h1>
      <!-- PRODUCT INPUTS -->
      <input type="text" id="product1" placeholder="Product 1">
      <input type="number" id="quantity1" placeholder="Quantity">
      <input type="number" id="price1" placeholder="Unit Price">
      <input type="text" id="product2" placeholder="Product 2">
      <input type="number" id="quantity2" placeholder="Quantity">
      <input type="number" id="price2" placeholder="Unit Price">
      <input type="text" id="product3" placeholder="Product 3">
      <input type="number" id="quantity3" placeholder="Quantity">
      <input type="number" id="price3" placeholder="Unit Price">
      <!-- ACTION BUTTONS -->
      <button onclick="calculateBill()">Calculate Bill</button>
      <button onclick="resetBill()">Reset</button>
      <!-- DISPLAY BILL -->
      <div id="result">Bill will appear here.</div>
   </div>
```

```html
    <!-- CONNECT JAVASCRIPT -->
    <script src="script.js"></script>
</body>
</html>
```

Style.css:

```css
/* ================= GENERAL STYLING ================= */
body {
    font-family: Arial;
    background: #eef7ef;
    text-align: center;
    padding: 30px;
}

/* ================= MAIN CONTAINER ================= */
.container {
    width: 450px;
    margin: auto;
    padding: 30px;
    background: white;
    border-radius: 12px;
}
/* ================= HEADING ================= */
h1 {
    color: #237a3b;
}
/* ================= INPUT FIELDS ================= */

input {
    width: 90%;
    padding: 10px;
    margin: 7px;
```

```
    border: 1px solid #aaa;
    border-radius: 5px;
}

/* ================== BUTTONS ================== */

button {
    padding: 10px 20px;
    margin: 10px 5px;
    background: #237a3b;
    color: white;
    border: none;
    border-radius: 5px;
    cursor: pointer;
}

/* ================== HOVER EFFECT ================== */

button:hover {
    background: #185b2b;
}
/* ================== BILL OUTPUT ================== */
#result {
    margin-top: 20px;
    padding: 15px;
    background: #f1f8f2;
    line-height: 1.8;
}
```

Script.js:

```
// ================== CALCULATE PRODUCT AMOUNT ==================
function calculateAmount(quantity, price) {
    // MULTIPLY QUANTITY AND PRICE
```

```
    return quantity * price;
}
// ================= CALCULATE BILL =================
function calculateBill() {
    // GET PRODUCT DETAILS
    let q1 = Number(document.getElementById("quantity1").value);
    let q2 = Number(document.getElementById("quantity2").value);
    let q3 = Number(document.getElementById("quantity3").value);
    let p1 = Number(document.getElementById("price1").value);
    let p2 = Number(document.getElementById("price2").value);
    let p3 = Number(document.getElementById("price3").value);
    // CALCULATE INDIVIDUAL AMOUNTS
    let a1 = calculateAmount(q1, p1);
    let a2 = calculateAmount(q2, p2);
    let a3 = calculateAmount(q3, p3);
    // CALCULATE TOTAL
    let total = a1 + a2 + a3;
    // APPLY DISCOUNT
    let discount = 0;
    if (total > 2000) {
        discount = total * 0.10;
    }
    // CALCULATE FINAL AMOUNT
    let finalAmount = total - discount;
    // DISPLAY RESULT
    document.getElementById("result").innerHTML =
        "Total Bill: ₹" + total.toFixed(2) + "<br>" +
        "Discount: ₹" + discount.toFixed(2) + "<br>" +
        "Final Payable Amount: ₹" +
        finalAmount.toFixed(2);
}
```

```
// ================= RESET FUNCTION =================
function resetBill() {
   // CLEAR ALL INPUTS
   document.querySelectorAll("input").forEach(input => {
      input.value = "";
   });
   // CLEAR RESULT
   document.getElementById("result").innerHTML =
      "Bill will appear here.";
}
```

Output:

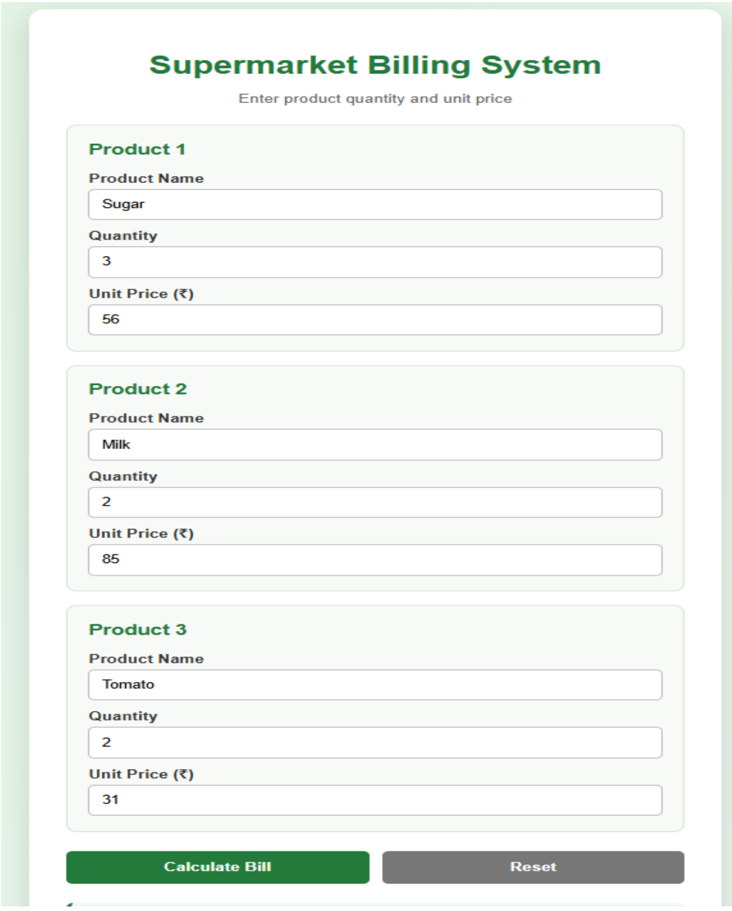


Figure no : 15 – Supermarket Billing

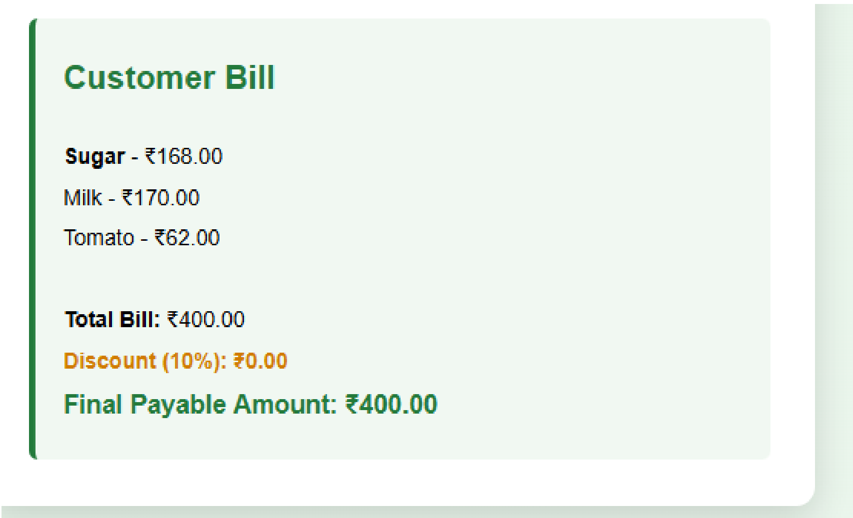


Figure no : 16 – Customer Bill

Explanation of Code:

- The **HTML** section accepts product names, quantities and unit prices.
- The **CSS** section styles the billing form, buttons and bill output.
- The calculateAmount() function calculates the amount for each product.
- Arithmetic operators calculate the **total bill**.
- An if statement applies a **10% discount when the total exceeds ₹2000**.
- The final payable amount is calculated after subtracting the discount.
- The resetBill() function clears all entered details.

Conclusion

The Supermarket Billing System demonstrates **HTML, CSS and JavaScript** concepts such as variables, operators, conditional statements and user-defined functions. It calculates the total bill, applies the required discount and displays the final payable amount.